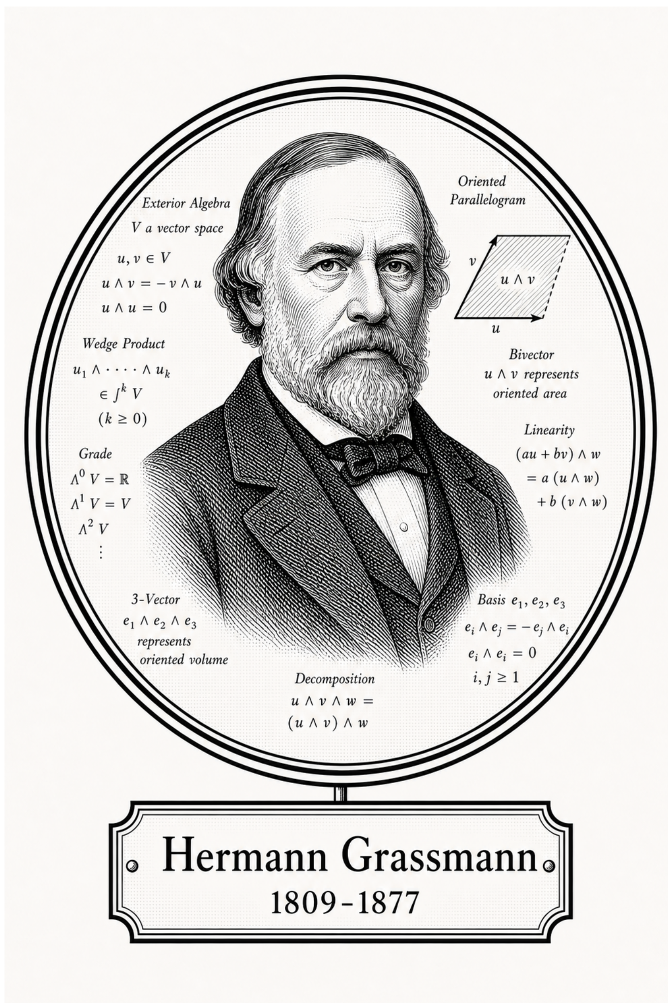
From Cantor to Ito

Algebra

Linear Algebra



Hermann Grassmann

1809-1877

Self-Study Notes

William Sollers

July 1, 2026

For every reader learning to make mathematics their own.

Contents

Chapter 1

Linear Algebra

1.1 Vector Space Preliminaries

Remark (Toolkit — Vector Space Preliminaries). **Concept family:** The concrete foundations for $\mathbf{F}^n$ before the abstract vector space axioms are introduced.

Atomic items in this section:

- Definition: List
- Definition: $\mathbf{F}^n$
- Definition: Addition in $\mathbf{F}^n$
- Theorem: Addition Is Commutative in $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$
- Definition: Additive Inverse in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Key predicate:

$$\text{ListEquality}((x_1,\dots,x_n),(y_1,\dots,y_m)) \;:\equiv\; n=m \,\wedge\, \forall k\le n\ (x_k=y_k).$$

Definition (List)

Definition 1.1.1 (List)**.** Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Remark (Standard quantified statement).

$$\mathrm{List}(L) \iff \exists n \in \mathbb{N} \cup \{0\} : \big(\mathrm{Length}(L) = n \wedge L = (x_1, x_2, \ldots, x_n)\big)$$

Remark (Definition predicate reading).

$$\mathrm{List}(L) \iff \exists n \in \mathbb{N}_0, \exists (x_1, \ldots, x_n) : L = (x_1, \ldots, x_n)$$

Remark (Interpretation). A list is the primary mechanism for discussing ordered finite sets of vectors or scalars. Unlike a set, the structural integrity of a list depends on both the identity and the position of its members.

- **Order Sensitivity**: The lists (a, b) and (b, a) are distinct unless $a = b$.
- **Finiteness**: By Axler's convention, a list must have a finite number of elements. Infinite ordered collections are categorized as sequences.
- **Empty List**: A list of length $n = 0$ is called the empty list, denoted by $()$.

Remark (Dependencies).

- Function
- Whole Numbers

Definition ($\mathbf{F}^n$)

Definition 1.1.2 ($\mathbf{F}^n$)**.** Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Remark (Standard quantified statement).

$$\begin{aligned} & x \in \mathbf{F}^n \\ \iff \quad & \exists x_1, \ldots, x_n \in \mathbf{F} \text{ such that } x = (x_1, \ldots, x_n). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Fn}(\mathbf{F}, n) \;:\equiv\; \mathbf{F}^n = \{(x_1, \ldots, x_n) : x_k \in \mathbf{F}\}.$$

Remark (Interpretation). An element of $\mathbf{F}^n$ is an ordered list of n scalars from $\mathbf{F}$. The order matters, and the list is completely specified by its coordinates.

Remark (Dependencies).

- Definition: List

Definition (Addition in $\mathbf{F}^n$)

Definition 1.1.3 (Addition in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\forall x, y \in \mathbf{F}^n,\\ &\quad x = (x_1, \ldots, x_n) \wedge y = (y_1, \ldots, y_n) \;\Rightarrow\; x + y = (x_1 + y_1, \ldots, x_n + y_n).\end{aligned}$$

Remark (Interpretation)*.* Addition in $\mathbf{F}^n$ is defined coordinatewise: add the first coordinates, then the second coordinates, and so on through the nth coordinates.

Remark (Dependencies)*.*

- Definition: $\mathbf{F}^n$

Theorem (Addition Is Commutative in $\mathbf{F}^n$)

Theorem 1.1.4 (Addition Is Commutative in $\mathbf{F}^n$)**.** *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall x, y \in \mathbf{F}^n,\; x + y = y + x.$$

Remark (Interpretation)*.* Because addition in $\mathbf{F}^n$ is coordinatewise and scalar addition in $\mathbf{F}$ is commutative, vector addition in $\mathbf{F}^n$ is commutative.

Remark (Dependencies)*.*

- Definition: Addition in $\mathbf{F}^n$

Definition (Zero Vector in $\mathbf{F}^n$)

Definition 1.1.5 (Zero Vector in $\mathbf{F}^n$)**.** The symbol 0 denotes the list of length n all of whose coordinates are 0:

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Standard quantified statement)*.*

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Interpretation). The zero vector is the vector whose every coordinate is the scalar 0 in $\mathbf{F}$. It is the additive identity for $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Definition (Additive Inverse in $\mathbf{F}^n$)

Definition 1.1.6 (Additive Inverse in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \ldots, -x_n).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\forall x \in \mathbf{F}^n,\\ &\quad x = (x_1, \ldots, x_n) \;\Rightarrow\; -x = (-x_1, \ldots, -x_n).\end{aligned}$$

Remark (Interpretation). The additive inverse in $\mathbf{F}^n$ is formed coordinatewise by taking the additive inverse of each coordinate in $\mathbf{F}$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$

Definition (Scalar Multiplication in $\mathbf{F}^n$)

Definition 1.1.7 (Scalar Multiplication in $\mathbf{F}^n$)**.** Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\forall \lambda \in \mathbf{F}\ \forall x \in \mathbf{F}^n,\\ &\quad x = (x_1, \ldots, x_n) \;\Rightarrow\; \lambda x = (\lambda x_1, \ldots, \lambda x_n).\end{aligned}$$

Remark (Interpretation). Scalar multiplication in $\mathbf{F}^n$ is defined coordinatewise: multiply each coordinate of the vector by the scalar λ.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Remark (Toolkit — Vector Spaces). **Concept family:** The abstract vector space axioms and their immediate consequences.

Atomic items in this section:

- Definition: Vector Space (10 axioms)
- Definition: Vector and Point
- Definition: Real Vector Space
- Definition: Complex Vector Space
- Theorem: Unique Additive Identity
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$
- Theorem: $a0 = 0$
- Theorem: $(-1)v = -v$

Key predicate:

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot).$$

The 10 axioms are closure, commutativity, associativity, additive identity, additive inverse, scalar closure, both distributive laws, scalar associativity, and the scalar identity.

Definition (Vector Space)

Definition 1.1.8 (Vector Space)**.** A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \to V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

(i) $u + v \in V$;

(ii) $u + v = v + u$;

(iii) $(u + v) + w = u + (v + w)$;

(iv) there exists $0 \in V$ such that $v + 0 = v$;

(v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;

(vi) $av \in V$;

(vii) $a(u + v) = au + av$;

(viii) $(a + b)v = av + bv$;

(ix) $a(bv) = (ab)v$;

(x) $1v = v$.

Remark (Standard quantified statement).

$$\begin{aligned}
&V \text{ is a vector space over } \mathbf{F} \\
&\iff \quad V \text{ is closed under addition and scalar multiplication,} \\
&\quad \text{addition on } V \text{ is commutative and associative,} \\
&\quad V \text{ has an additive identity and additive inverses,} \\
&\quad \text{and scalar multiplication satisfies the two distributive laws,} \\
&\quad \text{the associative law for scalar multiplication, and the identity law.}
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot)$$

Remark (Negated quantified statement).

$$\neg\ \mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot) \iff \text{at least one of the ten axioms fails for some } u, v, w \in V,\ a, b \in \mathbf{F}.$$ ■

Remark (Interpretation). A vector space is a set whose elements can be added together and multiplied by scalars, with these operations satisfying the natural algebraic rules familiar from $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$ (motivating example)
- Definition: Addition in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Definition (Vector and Point)

Definition 1.1.9 (Vector and Point)**.** An element of a vector space V is called a **vector**. When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Remark (Standard quantified statement).

$$v \in V \implies v \text{ is a vector},$$
$$x \in \mathbf{F}^n \implies x \text{ is a point of } \mathbf{F}^n.$$

Remark (Interpretation). The word *vector* emphasizes the algebraic role of an element of V. In $\mathbf{F}^n$, the same object can also be viewed geometrically as a point with coordinates in $\mathbf{F}$.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Real Vector Space)

Definition 1.1.10 (Real Vector Space)**.** A **real vector space** is a vector space over $\mathbb{R}$.

Remark (Standard quantified statement).

$$V \text{ is a real vector space} \iff V \text{ is a vector space over } \mathbb{R}.$$

Remark (Definition predicate reading).

$$\mathrm{RealVectorSpace}(V) \;:\equiv\; \mathrm{VectorSpace}(V, \mathbb{R}, +, \cdot).$$

Remark (Interpretation). In a real vector space, the scalars used in scalar multiplication are real numbers.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Complex Vector Space)

Definition 1.1.11 (Complex Vector Space)**.** A **complex vector space** is a vector space over $\mathbb{C}$.

Remark (Standard quantified statement).

$$V \text{ is a complex vector space } \iff V \text{ is a vector space over } \mathbb{C}.$$

Remark (Definition predicate reading).

$$\mathrm{ComplexVectorSpace}(V) \;:\equiv\; \mathrm{VectorSpace}(V, \mathbb{C}, +, \cdot).$$

Remark (Interpretation). In a complex vector space, the scalars used in scalar multiplication are complex numbers.

Remark (Notation). For each positive integer m, the notation $\mathbf{F}^m$ denotes the vector space of all lists of length m of elements of $\mathbf{F}$.

The notation $\mathbf{F}^\infty$ denotes the set of all infinite lists

$$(x_1, x_2, x_3, \dots)$$

with each $x_j \in \mathbf{F}$, equipped with coordinatewise addition and coordinatewise scalar multiplication.

Remark (Dependencies).

- Definition: Vector Space

Remark (Coordinates in Vector Spaces). A pointwise proof is available when vectors are given explicitly by coordinates, as in $\mathbf{F}^n$, because the operations are defined coordinatewise. In an arbitrary vector space, vectors need not come with coordinates, so proofs must use the vector space axioms rather than coordinate calculations.

Theorem (Unique Additive Identity)

Theorem 1.1.12 (Unique Additive Identity). *A vector space has a unique additive identity. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} &\text{If } 0, 0' \in V \text{ each satisfy} \\ &\qquad \forall v \in V,\ v + 0 = v \quad \text{and} \quad \forall v \in V,\ v + 0' = v, \\ &\text{then } 0 = 0'. \end{aligned}$$

Remark (Interpretation). Although the vector space axioms assert the existence of an additive identity, they force that identity to be unique.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Unique Additive Inverse)

Theorem 1.1.13 (Unique Additive Inverse). *Every vector in a vector space has a unique additive inverse. Go to proof.*

Remark (Standard quantified statement).

$$\forall v \in V,\ \text{if } w, u \in V \text{ satisfy } v + w = 0 \text{ and } v + u = 0,$$
$$\text{then } w = u.$$

Remark (Interpretation). The vector space axioms guarantee that each vector has an additive inverse, and that inverse is determined uniquely by the requirement that the sum be 0.

Remark (Notation). If $v \in V$, then the unique additive inverse of v is denoted by $-v$.

If $w, v \in V$, then

$$w - v \ := \ w + (-v).$$

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Identity

Theorem $(0v = 0)$

Theorem 1.1.14 $(0v = 0)$**.** *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ 0v = 0.$$

Remark (Interpretation). Multiplying any vector by the scalar 0 gives the zero vector. The 0 on the left is the scalar zero in $\mathbf{F}$; the 0 on the right is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem $(a0 = 0)$

Theorem 1.1.15 $(a0 = 0)$**.** *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbf{F},\ a0 = 0.$$

Remark (Interpretation). Every scalar sends the zero vector to the zero vector. The 0 on the right of a is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem $((-1)v = -v)$

Theorem 1.1.16 $((-1)v = -v)$**.** *For every* $v \in V$,

$$(-1)v = -v.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ (-1)v = -v.$$

Remark (Interpretation). Scalar multiplication by -1 produces the additive inverse of a vector. This justifies the notation $-v$ for the additive inverse: it is literally (-1) times v.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$

Remark (Toolkit — Vector Subspaces). **Concept family:** Subspaces, their sums, and direct sums.

Atomic items in this section:

- Definition: Subspace
- Theorem: Conditions for a Subspace
- Definition: Sum of Subspaces
- Theorem: Sum of Subspaces Is the Smallest Containing Subspace
- Definition: Direct Sum
- Theorem: Condition for a Direct Sum
- Theorem: Direct Sum of Two Subspaces

Key predicates:

Subspace(U, V), ClosedUnderAddition(U), ClosedUnderScalarMultiplication$(U, \mathbf{F})$, DirectSu

Key test: U is a subspace $\iff$ $0 \in U$, U closed under $+$, U closed under scalar multiplication.

Definition (Subspace)

Definition 1.1.17 (Subspace)**.** Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\iff \quad U \subseteq V \text{ and } U \text{ is a vector space over } \mathbf{F} \\ &\quad \text{under the operations inherited from } V. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Subspace}(U, V) \;:\equiv\; U \subseteq V \;\wedge\; \mathrm{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Subspace}(U, V) \iff U \not\subseteq V \;\vee\; \neg\,\mathrm{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Interpretation). A subspace is a subset of a vector space that is closed under the vector-space operations and therefore forms a vector space in its own right.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Conditions for a Subspace)

Theorem 1.1.18 (Conditions for a Subspace)**.** *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$*;*

(ii) for all $u, v \in U$*, one has* $u + v \in U$*;*

(iii) for all $a \in \mathbf{F}$ *and all* $u \in U$*, one has* $au \in U$*.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\iff \quad 0 \in U \\ &\quad \wedge\; \mathrm{ClosedUnderAddition}(U) \\ &\quad \wedge\; \mathrm{ClosedUnderScalarMultiplication}(U, \mathbf{F}). \end{aligned}$$

Remark (Failure modes). U fails to be a subspace via exactly one of three routes: it omits the zero vector, it contains two vectors whose sum falls outside U, or it contains a vector and a scalar whose product falls outside U.

Remark (Contrapositive quantified statement).

$$\neg\, \mathrm{Subspace}(U, V) \iff 0 \notin U \vee \neg\, \mathrm{ClosedUnderAddition}(U) \vee \neg\, \mathrm{ClosedUnderScalarMultiplication}(U, \mathbf{F}).$$

Remark (Interpretation). To check that a subset is a subspace, it suffices to verify the three conditions. The remaining seven vector space axioms are inherited automatically from V since U uses the same operations. The three conditions are minimal: condition (i) ensures U is non-empty; (ii) and (iii) ensure the inherited operations stay within U.

Remark (Dependencies).

- Definition: Subspace
- Definition: Vector Space

Definition (Sum of Subspaces)

Definition 1.1.19 (Sum of Subspaces)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. The **sum** of $U_1, \ldots, U_m$ is the set

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \ldots, u_m \in U_m\}.$$

Remark (Standard quantified statement).

$$\begin{aligned} & v \in U_1 + \cdots + U_m \\ & \quad \iff \quad \exists u_1 \in U_1 \ \cdots \ \exists u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{SumOfSubspaces}(W, \{U_1, \ldots, U_m\}, V) \;:\equiv\; W = U_1 + \cdots + U_m.$$

Remark (Interpretation). The sum of subspaces consists of all vectors that can be built by adding together one vector from each subspace. Vectors from different subspaces interact; the representation $v = u_1 + \cdots + u_m$ need not be unique.

Remark (Dependencies).

- Definition: Subspace

Theorem (Sum of Subspaces Is the Smallest Containing Subspace)

Theorem 1.1.20 (Sum of Subspaces Is the Smallest Containing Subspace)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \ldots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \ldots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\operatorname{Subspace}(U_1 + \cdots + U_m,\, V),\\
&U_j \subseteq U_1 + \cdots + U_m \quad \text{for } j = 1, \ldots, m,\\
&\forall W \subseteq V \; \Big[\big(W \text{ is a subspace of } V \wedge U_1 \subseteq W \wedge \cdots \wedge U_m \subseteq W \big)\\
&\qquad\qquad \Rightarrow \; U_1 + \cdots + U_m \subseteq W \Big].
\end{aligned}
$$

Remark (Interpretation). The sum is the smallest subspace containing all the given subspaces: it is their join in the lattice of subspaces of V. Any subspace containing all of $U_1, \ldots, U_m$ must contain every sum $u_1 + \cdots + u_m$, so it contains the sum subspace.

Remark (Dependencies).

- Definition: Sum of Subspaces
- Theorem: Conditions for a Subspace

Definition (Direct Sum)

Definition 1.1.21 (Direct Sum)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V such that

$$V = U_1 + \cdots + U_m.$$

Then V is called the **direct sum** of $U_1, \ldots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \cdots + u_m, \qquad u_1 \in U_1, \ldots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \cdots \oplus U_m.$$

Remark (Standard quantified statement).

$$
\begin{aligned}
&V = U_1 \oplus \cdots \oplus U_m\\
&\quad \iff \quad V = U_1 + \cdots + U_m\\
&\qquad \wedge \; \forall v \in V \; \exists!\, u_1 \in U_1 \; \cdots \; \exists!\, u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m.
\end{aligned}
$$

Remark (Definition predicate reading).

$$\operatorname{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \;:\equiv\; V = U_1 \oplus \cdots \oplus U_m.$$

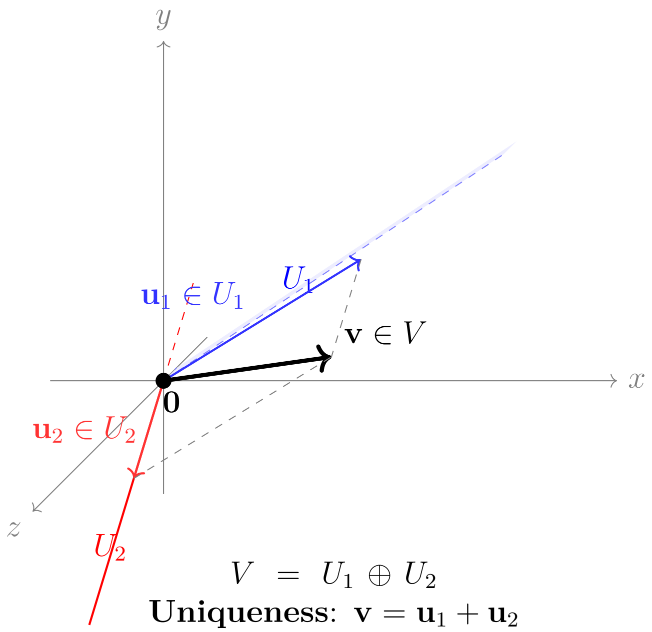
Remark (Negated quantified statement).

$$\neg\, \operatorname{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \iff \exists v \in V \; \exists \text{ two distinct decompositions of } v.$$

Remark (Interpretation). A direct sum is a sum in which every vector has a unique decomposition into pieces coming from the given subspaces. The uniqueness condition is the structural rigidity that makes direct sums so useful — it means the subspaces contribute independently, with no redundancy.

Remark (Dependencies).

- Definition: Sum of Subspaces



$V = U_1 \oplus U_2$
Uniqueness: $\mathbf{v} = \mathbf{u}_1 + \mathbf{u}_2$
Trivial Intersection: $U_1 \cap U_2 = \{\mathbf{0}\}$

Theorem (Condition for a Direct Sum)

Theorem 1.1.22 (Condition for a Direct Sum)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$*;*

(ii) if $u_1 \in U_1, \dots, u_m \in U_m$ *and*

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &V = U_1 \oplus \cdots \oplus U_m \\ &\quad \iff \quad \forall u_1 \in U_1 \ \cdots \ \forall u_m \in U_m \\ &\qquad \big(u_1 + \cdots + u_m = 0 \Rightarrow u_1 = \cdots = u_m = 0\big). \end{aligned}$$

Remark (Interpretation). A sum is direct exactly when the zero vector has only the trivial decomposition as a sum of vectors from the given subspaces. This is the practical test: rather than verifying uniqueness for every $v \in V$, it suffices to check uniqueness at $v = 0$.

Remark (Dependencies).

- Definition: Direct Sum
- Theorem: Unique Additive Identity

Theorem (Direct Sum of Two Subspaces)

Theorem 1.1.23 (Direct Sum of Two Subspaces)**.** *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U + W = U \oplus W \\ &\iff \quad U \cap W = \{0\}. \end{aligned}$$

Remark (Contrapositive quantified statement).

$$U \cap W \neq \{0\} \iff U + W \neq U \oplus W.$$

If the two subspaces share a nonzero vector, the sum is not direct.

Remark (Interpretation). For two subspaces, directness is equivalent to saying they overlap only in the zero vector. This gives a simple geometric test: draw both subspaces and check whether their intersection is just the origin. The result does not generalise to three or more subspaces — it is possible for $U_1 \cap U_2$, $U_1 \cap U_3$, and $U_2 \cap U_3$ to all equal $\{0\}$ yet $U_1 + U_2 + U_3$ still fail to be a direct sum.

Remark (Dependencies).

- Theorem: Condition for a Direct Sum
- Definition: Direct Sum

Bibliography